MD. RAKIB HOSSAIN

Machine Learning · Computer Vision · AI Automation
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PROFESSIONAL SUMMARY

Final-year CSE student at AIUB and IT Intern at Live Technologies who builds machine-learning systems end to end: collecting and labelling data, training and calibrating models, and shipping them inside web apps people use. First author of three papers presented at IEEE COMPAS 2026 and ICCA 2026. Working toward AI and automation engineering; uses AI-assisted development (Claude Code) daily.

EXPERIENCE

IT Intern, Live Technologies Ltd, Dhaka | Oct 2026 – Jan 2027
- Three-month internship in the Department of Information Technologies, working on data analysis, machine-learning and AI projects, backend and API development, and database integration.

Full-Stack Developer (Freelance), Durbar Robotics | Oct 2026 – Present
- Building durbarrobotics.com in Next.js: an e-commerce store for robotics parts, a course platform with video lessons, and an instant 3D-printing quote tool that accepts STL/OBJ/STEP uploads.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, C++
ML & Data: PyTorch, scikit-learn, pandas, NumPy, CatBoost, XGBoost, YOLO, OpenCV, SHAP
Methods: Model calibration, conformal prediction, leakage auditing, feature engineering, benchmarking, experiment design
Apps & Tools: FastAPI, NestJS, Next.js, React, PostgreSQL, Git/GitHub, Vercel, Render, Arduino/ESP32

PROJECTS AND PUBLICATIONS

RideGuard – Calibrated, Auditable Crash-Risk Model and Web App | *Python*, *CatBoost*, *SHAP*, *FastAPI*, *NestJS*, *Next.js* | Research | App Code | Live
- Built a calibrated CatBoost risk model for Dhaka motorcyclists: isotonic recalibration (ECE 0.043 → 0.008), conformal prediction sets (89.5% coverage at a 90% target), ADWIN drift detection and SHAP explanations.
- Its self-audit showed the data carried little real pre-crash signal, so I reported that, rather than the accuracy, as the central finding. Presented at IEEE COMPAS 2026 (first author).
- Shipped it as a web app: a FastAPI service serves the model to a NestJS API and a Next.js front end with a live risk map, route risk and voice alerts.

DhakaNight – Night-Time Object Detection Dataset and Benchmark | *PyTorch*, *YOLOv8*, *OpenCV*, *Roboflow* | Dataset | Code
- Built and labelled a 2,380-image dataset of night-time Dhaka traffic with 45,224 boxes across 10 classes, including CNGs, rickshaws and human haulers.
- Benchmarked CLAHE, Gamma, Zero-DCE and RetinexFormer against a YOLOv8 baseline: none improved detection, while a larger detector gave the best mAP@0.5 (0.610 vs 0.587). Presented at ICCA 2026 (first author).

Drikon – Full-Stack E-Commerce Platform with Recommendation Engine | *Next.js*, *NestJS*, *PostgreSQL*, *Prisma* | Live | Code
- "Frequently bought together" and personalised recommendations from a from-scratch Apriori implementation over real order history, scored by support, confidence and lift.
- 17 backend modules with secure accounts (2FA, role-based access), voice search, product comparison, and Bangla/English localisation.

PulseStone – Handheld Anxiety-Relief Device | *ESP32-C3*, *Arduino C++*, *Python* | Code
- One squeeze starts a guided breathing exercise. In a 15-participant, 150-trial study, users started in 0.52 s versus 7.49 s on a phone app (14.5× faster), and all 15 preferred it. Presented at ICCA 2026 (first author).

LEADERSHIP AND AWARDS

Founder and Team Lead, "Khepa Chakka" Robotics Team | 2023 – Present
- Lead a competitive team building line-following, soccer, race and battle robots across mechanical design, electronics and software.
- Winner, Cozmo Clench – Techfest IIT Bombay (Bangladesh Zonal), 2024 · Champion, Robot Soccer – AIUB CS Fest, 2024 · National Robotics Championship 2025: 1st Runner-up, Nano Battle Bot; 2nd Runner-up, Robo Race · 1st Runner-up, Robo Race – Technoxian Bangladesh, 2024 · 1st Runner-up, Robot Soccer – AIUB Robotic Crew (ARC), 2025

Volunteer, Bangladesh ICT & Innovation Awards 2026 | Certificate of appreciation from the Bangladesh ICT and Innovation Network (BIIN)

EDUCATION

B.Sc. Computer Science and Engineering, American International University-Bangladesh (AIUB) | CGPA 3.30/4.00 | Expected 2026
Thesis: Accuracy Analysis of Hybrid UTDoA/AoA Positioning in Sparse Rural Road Networks
Higher Secondary Certificate (HSC), Dhaka Imperial College, 2021 | Secondary School Certificate (SSC), Dhaka Collegiate School, 2019
Languages: Bangla (Native), English (Intermediate)

REFERENCE

Dr. Muhammad Hasibur Rashid Chayon | Associate Professor, Department of Computer Science, Faculty of Science and Technology
American International University-Bangladesh (AIUB) | Contact details available on request